

Standard Operating Procedure (SOP): Cold Storage Management

This Standard Operating Procedure (SOP) defines the mandatory processes for receiving, storing, monitoring, and maintaining temperature-sensitive products within cold storage units to ensure product safety, quality, and regulatory compliance.

1. Purpose and Scope

- **Purpose:** Ensure safe handling, storage, and preservation of temperature-sensitive goods (such as food, produce, or pharmaceuticals).
- **Scope:** Applies to all personnel responsible for handling goods in walk-in coolers, freezers, or cold rooms.

2. Receiving and Storing Goods

- **Immediate Transfer:** Move perishable or temperature-sensitive items directly into the designated cold room or cooler upon arrival to prevent thermal degradation.
- **Elevation & Clearance:** Store all boxes and pallets off the floor on clean racks. Maintain at least a 2-inch gap from walls and 6 inches from ceilings or vents to allow unobstructed air circulation.
- **Segregation:** Store non-food items below food, and allergens below non-allergens. Ensure raw meats are stored separately or below ready-to-eat items.
- **Inventory Rotation:** Implement the First-In, First-Out (FIFO) method. Clearly label batches with item name, entry date, and expiration date.

3. Temperature and Humidity Specifications

Product Type	Temperature Range	Relative Humidity
Fresh Produce	0°C to 4°C (32°F to 39°F)	85% – 95%
Dairy & Pharmaceuticals	2°C to 8°C (36°F to 46°F)	Ambient / Controlled
Meat & Seafood (Frozen)	-18°C (0°F) or lower	N/A

- **Monitoring:** Manually verify and record internal temperatures at least twice daily (morning and evening) or utilize automated continuous logging systems.

4. Sanitation and Maintenance

- **Cleaning Schedules:** Regularly inspect and sanitize the interior to remove spills, ice buildup, debris, or expired inventory.
- **Infrastructure Integrity:** Regularly inspect door gaskets, strip curtains, floor seals, and insulation panels to eliminate cool-air leaks.

- **Preventative Maintenance:** Schedule certified technicians for routine maintenance of compressors, condensers, evaporators, and automatic defrost cycles.

5. Temperature Deviations and Contingencies

- **Quarantine Protocol:** If temperatures fall outside authorized safe limits, immediately quarantine affected batches and label them 'Do Not Use' pending quality control review.
- **Emergency Alerts:** Maintain operational backup generators and continuous telemetry systems configured to send immediate SMS/email alerts during power or system failures.
- **Documentation:** Log all excursions, root cause analyses, and completed corrective and preventive actions (CAPA) in the facility registry.